

A functional description of SDH transmission equipment

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For some time, functional decomposition techniques have been applied to describe the architecture of telecommunications transport networks. A group within the transmission and multiplexing (TM) sector of the European Telecommunications Standards Institute (ETSI) has now applied the technique to provide a rigorous means of describing telecommunications transmission equipment. This paper describes the application of the technique to equipment conforming to the ITU-T Recommendations for a synchronous digital hierarchy (SDH), and describes the benefits of the technique to both equipment manufacturers and network operators.

1. Introduction

This paper provides a guide to the principles behind, and the usage of, European Telecommunications Standard (ETS) 300 417 [1] which describes the 'Generic Functional Requirements for Synchronous Digital Hierarchy Transmission Equipment'. The standard is the first to apply functional decomposition techniques to the specification of telecommunications equipment. The methodology will be applied more widely in the future as organisations seek to move away from the current practice of producing international standards for specific items of equipment.

This introduction to the standard is intended to air wider discussion of the issues involved and to act as a reference for those using the standard. The paper is based on the current issue of the ETS, contributions to experts group meetings, and draft versions of tutorial material.

2. The need for standardisation

In the past, certain manufacturers of telecommunications equipment may have enjoyed secure home markets for their products and developed a close relationship with their home telecommunications authority. This enabled the supplier to offer equipment designed to meet the specific needs of these operators. In these days of greater competition, home markets are less secure and suppliers are forced to aggressively market their products to other operators.

The synchronous digital hierarchy (SDH) has been designed to facilitate interoperation of equipment produced by different manufacturers which further widens the competitive aspects of equipment purchase. A supplier of SDH equipment may therefore supply a number of operators, possibly within a single country, and may be presented with a number of different equipment

specifications. At best, this leads to duplication of effort with several, often very comprehensive, specifications relating to the same piece of equipment. In many cases, especially in a period of standards evolution, the specifications each require slightly different functionality which may restrict access by certain manufacturers to key markets.

From the operator's point of view, the necessity to fully specify an item of SDH equipment in order to avoid confusion and misinterpretation of technical requirements by a number of different manufacturers has led to increased specification complexity, which has made it difficult to judge between competing suppliers. Adopting a common methodology for describing SDH equipment is therefore intended to simplify the specification process, to ensure fair competition and to prevent misunderstanding. It should also present a set of common basic equipment requirements, facilitating interoperation of multivendor equipment and driving down costs to both the operator and the end user.

3. SDH transmission within BT

Equipment conforming to the synchronous digital hierarchy will form the basis of the next generation of transport networks for both BT and its competitors.

BT has deployed two SDH networks using equipment from different manufacturers. One network supplied by GPT forms the basis of a core transmission network which is scheduled to cover the country by the end of 1997 as shown in Fig 1; the other, procured from Ericsson, addresses the specific requirements of the international backhaul network. Other network operators within the UK are also deploying SDH equipment supplied by a number of

manufacturers. Many of these competitors are now requesting direct SDH interconnection with BT which stretches the interoperability capabilities of SDH equipment to the limits.

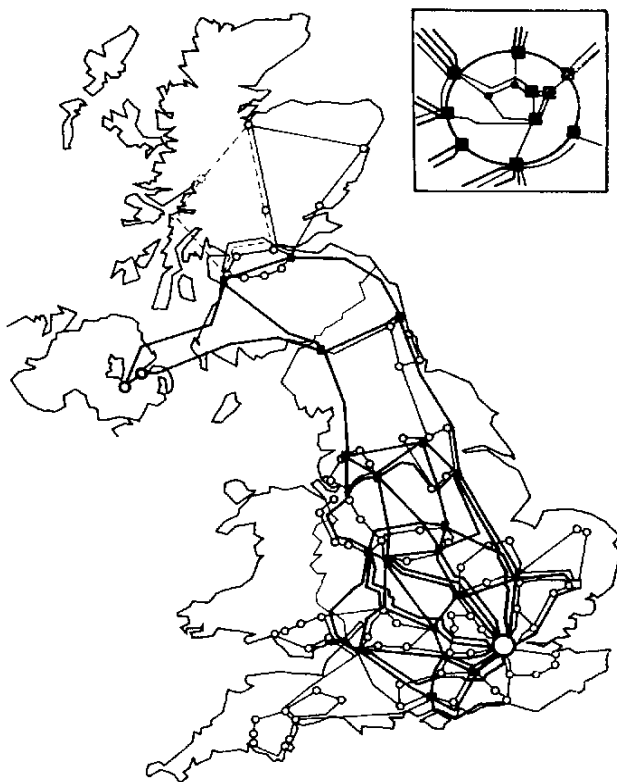


Fig 1 BT SDH network 1996/97.

In such an environment, an unambiguous specification methodology for equipment becomes essential. It is this function which the ETS seeks to perform.

4. The functional modelling technique

A telecommunications network may be viewed in many ways. For example, to a particular customer purchasing a service the network may be viewed as a simple pipe which transports the required information with the desired accuracy at an agreed price; to a certain extent the actual composition of the network is somewhat academic.

To the operator, however, the network is a complicated interaction of switching and transmission systems, diverse transmission media, signalling systems, and management capabilities. Furthermore, the network as presented to the customer should be a homogeneous entity, even though the global transmission network is composed of an interconnection of geographical regions each comprising the networks of several private or public companies.

In addition to the simple partitioning of networks according to ownership, networks may be decomposed on a service or hierarchical basis. This is the principle of network

'layering' which is the basis of the functional modelling approach first proposed by the ITU in Recommendation G.803 [2].

The example in Fig 2 illustrates the principle that a connection across one network layer may physically consist of a series of connections, often in other layers or in other domains. The specific example illustrates that an SDH VC-12 connection will in practice be carried within a VC-4, which in turn would be transported over a multiplex section (and regenerator section and physical section layers).

Such a description portrays any network as a series of hierarchical layers, each containing the logical functions of connection, termination and adaptation. A group within the European Telecommunications Standards Institute have now applied the technique to provide a methodology which can be used to describe telecommunications transmission equipment.

This has been done by rigorously defining the processes contained within these logical functions (called atomic functions, as they reflect the smallest 'sensible' partition of equipment), the interaction with common equipment functions, and the way in which functions may be combined.

Processes have been assigned to these functions in order to model the physical behaviour of an equipment as measured at an interface. In order to describe a particular item of transmission equipment it is necessary to define and to interconnect the functions of several layers. The SDH transmission layers and interface signals defined for the European hierarchy are illustrated in Fig 3.

Each layer constitutes a server for the hierarchical layers above it, while acting as a client for those below. A function may act as a server to auxiliary layers in addition to its main client. Other global layers exist which interface to most of the hierarchical functions. These include the synchronisation, management, and power supply. This model may be extended to include other layers into which actual services may be mapped.

ETS 300 417 seeks to apply the principles of functional modelling to the design and specification of SDH equipment. The actual behaviour of any SDH equipment can only be measured at physical interfaces. The modelling technique provides a convenient means of assigning aspects of physical behaviour to logical functions.

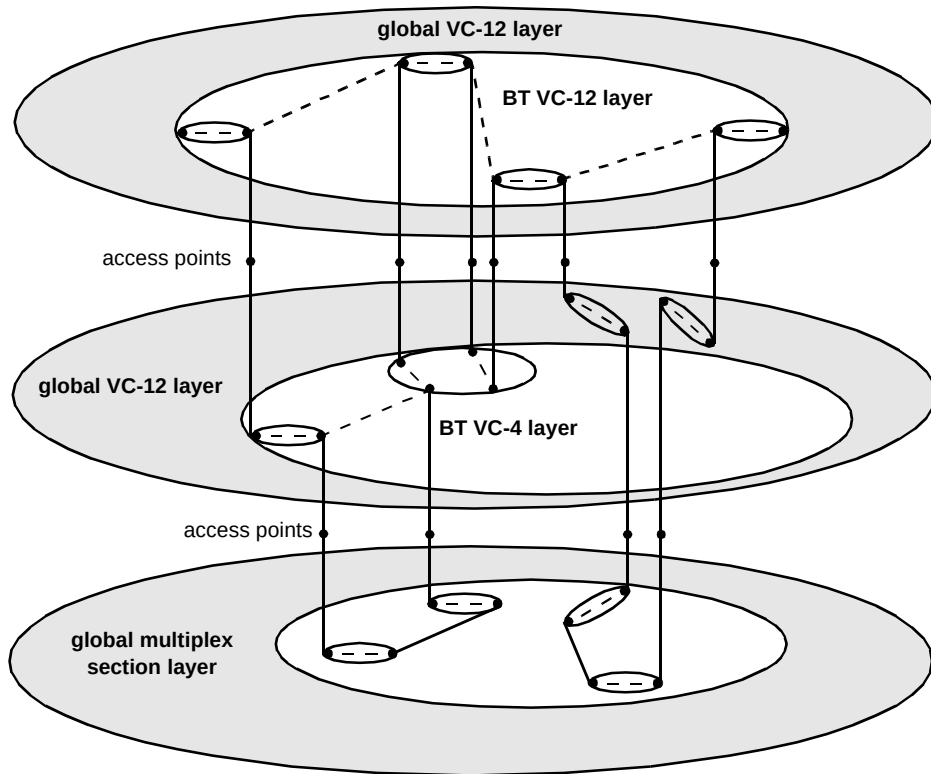


Fig 2 Example of SDH network decomposition.

5. ETS structure

ETS 300 417 is intended for use by the authors of telecommunications equipment specifications and equipment designers. It is structured to provide a useful reference document for the regular user. It does not contain any explanatory material which would be superfluous to such a user and is therefore intimidating to the new reader.

The ETS is structured to reflect the layered nature of the transmission hierarchy. It currently consists of eight parts which are in various stages of production and publication. The only part of the ETS to be published to date defines generic processes and performance objectives. It contains material on functional modelling, function combination, and defined sub-network functions such as loopbacks.

Other parts of the ETS define in great detail the specific atomic functions associated with the physical section layer, the STM-N regenerator and multiplex section layer, the SDH path layers, path layers for the existing plesiochronous digital hierarchy (PDH), and the synchronisation distribution layer. The final parts of the ETS define management systems and common groupings of functions.

The intention of the ETS is to provide a library of basic or 'atomic' functions for each layer, and to define the rules for combining functions to form logical descriptions of equipment.

5.1 Diagrammatic convention

Within the ETS, three types of atomic function have been defined — connection, adaptation, and termination. They are each represented by defined symbols as illustrated in Fig 4.

The connection function provides flexibility within a network layer and may either be configured by a management system, as for example in the case of virtual container (VC) assignment within an add-drop multiplexer (ADM), or autonomously as in the case of automatic protection switching.

The termination function processes the overheads associated with a particular layer and is responsible for such functions as performance monitoring.

The adaptation function adapts information from a client layer into a server layer and provides the conversion processes such as frame alignment and timing recovery.

Each function is present within each hierarchical layer of the network, although certain functions may be null and there may be more than one adaptation function in any layer. In order to simplify the diagrammatic representation of complex equipment functions, certain conventions have been adopted for combinations of functions. The functions may also be either unidirectional or bidirectional.

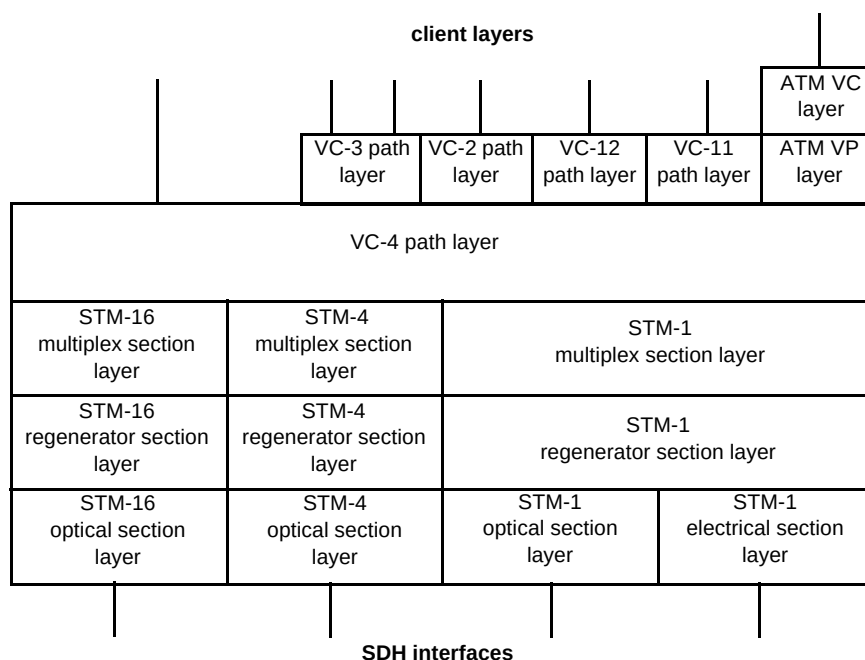


Fig 3 SDH network layers.

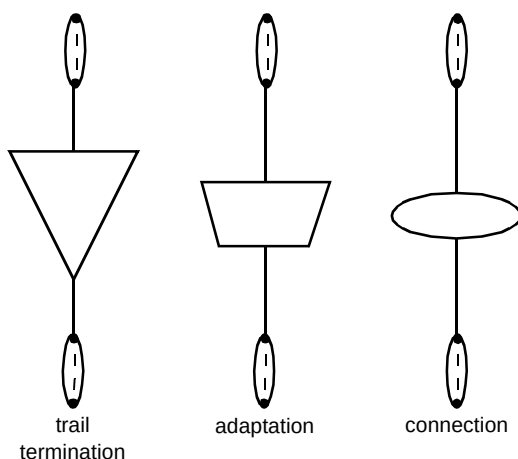


Fig 4 Diagrammatic convention.

5.2 Nomenclature

The adaptation functions are named according to their client and server layer. Therefore the function S4/S12_A bi-directionally adapts a client signal in the VC-12 layer into the VC-4 layer, and comprises both source (S4/S12_A_So) and sink (S4/S12_A_Sk) components. Because the definition is based on layers rather than on signal characteristics, the signal formats at the inputs and outputs of two different atomic functions may be identical.

The termination functions are named according to the logical layer which they terminate, e.g. S4_TT_So.

A nomenclature is also used to identify the information associated with a particular network layer or function. The signals within a layer can be either characteristic information (CI) associated with the layer, adapted information (AI) from the client layer, timing information (TI) or management information (MI). For a particular layer these are referred to as, for example, S4_CI, S4_AI, S4_TT_MI, S4_TI_CK. Adapted and characteristic information typically consists of data, clock, frame start reference and a signal fail indication. These are individually referred to as, for example, S4_CI_D or S4_AI_FS. Similarly, the management information may be subdivided into configuration commands, command responses, management system reports and performance monitoring parameters.

The processes within atomic functions generate defects, consequent actions, fault cause correlation information and performance monitoring information. These are all identified by a three or four letter acronym, common examples of which are:

dTIM	trail identifier mismatch
aTSF	trail signal fail
cSSF	server signal fail
pNEBC	near-end errored block count

These are fully described by reference to the function to which they belong. Hence the full description of the command to provision the path trace identifier transmitted by a VC-12 is S12_TT_So_MI_T_xTI, and the management

report for a VC-4 path trace mismatch is S4_TT_Sk_MI_cTIM.

6. Function definition

The ETS separates those functions which process client data, or associated signals such as synchronisation, from those which perform common management functions. The specification of the common management process is somewhat generic and only describes processes directly related to the client information. Two types of function are defined within the ETS for the processing of client data — atomic functions and compound functions. Atomic functions are the fundamental components of equipment specification, compound functions are commonly used groups of atomic functions.

6.1 Atomic functions

The definition of atomic functions within the ETS has been structured to form a library of parts which may be combined to provide a functional description of any SDH equipment. New functions may be added to the library at any time to reflect enhancements to the standards or the addition of new network layers.

Each function is described in a standard manner, thus reinforcing the library analogy. The description consists of:

- a textual description of the function,
- a definition of the input and output data, together with any associated auxiliary channel interfaces, under both normal and abnormal operating conditions,
- a description of the processes carried out by the function including any signal conditioning,
- a definition of any defects, such as loss of signal, which are detected by the function,

- details of any performance monitoring,
- any consequent actions performed upon the receipt of defect or anomaly declarations.

Each of these is defined in an explicit, unambiguous manner, thereby ensuring compatibility between different physical implementations of the same logical function. For example, the criteria for detecting a mismatch in the SDH access point identifier are defined in terms of format, acceptance, noise immunity and defect persistence.

Future issues of the ETS may also include a diagrammatic description of each atomic function which would aid understanding.

Typically, many of the processes would be distributed between the atomic functions and the common management function; the distinction being made on whether a process affects a specific trail or a layer. An example of this partitioning for the performance and fault management processes is illustrated in Fig 5. In this case the raw information is processed by the atomic function to produce both fault and performance information, each of which is passed to a different client within the equipment management function.

6.2 Compound functions

Any combination of functions that are normally used as a unit may be defined as a compound function. If the grouping transcends the boundaries between hierarchical layers the unit is termed a major compound function. The major compound functions defined in the ETS to date include the regenerated section termination (for use within terminal equipment of systems with regenerators where the section overhead is terminated), the unregenerated section termination (for systems without regenerators), and the elementary regenerated section termination (for use at regenerators where the multiplex section overhead is to be passed transparently).

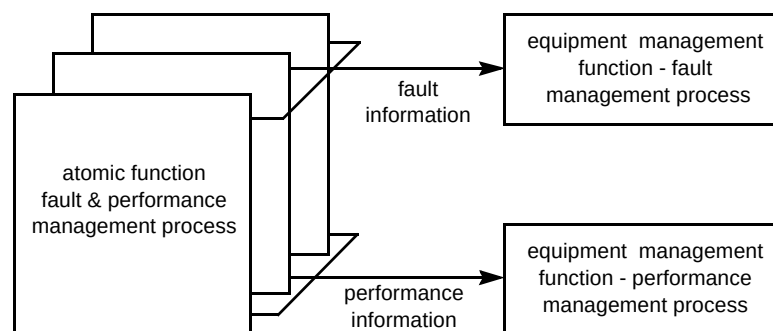


Fig 5 Equipment management functions.

7. Specific functions

Partitioning an SDH network element into hardware and software is somewhat arbitrary as the hardware may contain much embedded processing. The functional modelling technique assumes that all functions are logical and imposes no constraints on physical implementation. The only separation of functions is between those that directly affect the client data within a layer and those common functions which might typically be associated with an element manager.

ITU Recommendation G.803 describes the architectures of transport networks in terms of client and server layers. Each of these layers contains flexibility, adaptation of information between layers and termination of any information associated with the layer. ETSI have extended this principle to describe the generic functional requirements for SDH transmission equipment in ETS 300 417.

These functions are generally bidirectional, i.e. they contain both source and sink functions, and monitor the status of the information which passes through them. The fundamental basis for performance monitoring is the detection of anomalies, which are minor discrepancies between the received and expected information. These are converted into defects through the application of a persistency check. A certain amount of processing of these defects is performed by the atomic functions themselves but higher level correlation is performed by the element manager function.

The interfaces between the functions are via connection points (CP) which lie within a single layer, access points (AP) which connect a server layer to the adaptation function of the client layer, timing points (TP) across which timing information is exchanged with the equipment synchronisation layer, and management points (MP) across which information is exchanged with the management system. All of these are logical reference points, not physical interfaces. For example, the management points represent a logical interface between management functions and equipment functions — they do not represent a defined Q3 interface. The locations of these reference points are indicated in Fig 6.

The information which crosses an access point is termed adapted information as it has been adapted from the client layer. The information which crosses a connection point is termed characteristic information as it is specific to the server layer.

7.1 Adaptation functions

Adaptation functions process client layer information in such a way that it is matched to the requirements of the

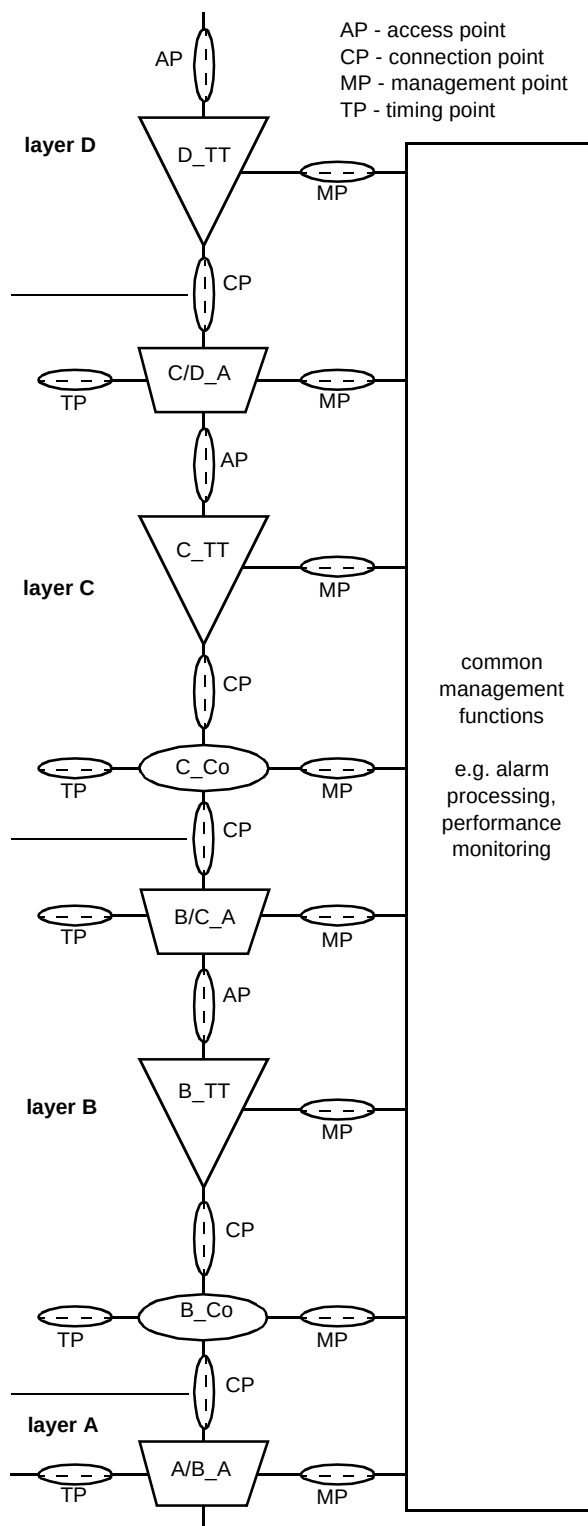


Fig 6 Functions within a network layer.

server layer, and hence they represent the conversion process between a server and a client layer. The following processes may be present in an adaptation function:

- scrambling/descrambling,
- encoding/decoding,

- alignment (framing, pointer interpretation),
- bit rate adaptation,
- frequency justification,
- multiplexing/demultiplexing,
- timing recovery,
- smoothing.

The scrambling process alters digital data in accordance with a predefined algorithm to ensure the resulting bit-stream has a sufficient density of transitions to allow bit clock recovery from it. The descrambling process recovers the original digital data from the scrambled bit-stream.

The encoding/decoding process conditions a digital data stream to the characteristics of the physical medium over which it is to be transported. The decoding process recovers the original digital data from the medium specific form in which it is received.

The alignment process may be specific to the client layer data. In the SDH environment the alignment process is typically one of frame or multi-frame detection or pointer interpretation. The defects are reported to the fault management process.

The bit rate adaptation process accepts input information at a certain bit rate and outputs that same information at a different bit rate. In the source direction, this process creates gaps in which other adaptation functions can add their signals. An example of this is the S12/P12_A_So function which maps a PDH 2048-kbit/s signal into an SDH container; the 2-Mbit/s signal input to this function is output at a higher bit rate. The created gaps are filled with the VC-12 POH. The frequency justification process is used to accommodate any frequency (and/or phase) differences between input and output signals. The commonly used terms mapping and demapping are covered by bit rate adaptation and frequency justification processes.

The multiplexing and demultiplexing processes, common in transmission equipment, are modelled by means of multiple adaptation functions, connected to one access point. In the source direction, the information applied by each connected adaptation function ends up in pre-allocated timeslots of the resulting time division multiplexed signal. Adaptation sink functions extract their associated adapted information from the common access point.

The timing recovery process extracts a clock signal from the incoming data signal. The timing recovery process is performed in the adaptation sink function in the physical section layer, e.g. in OS16/RS16_A_Sk. These functions also smooth the data to remove any phase steps.

Adaptation sink functions may monitor the following transmission defects:

- alarm indication signal,
- loss of pointer,
- loss of frame,
- loss of multiframe.

The generation of anomalies and defects are process specific and are described in the definitions of individual atomic functions within the ETS. However, the general process for the detection of anomalies, the declaration of defects, the control of consequent actions and the identification of fault cause within the atomic adaptation function are illustrated in Fig 7.

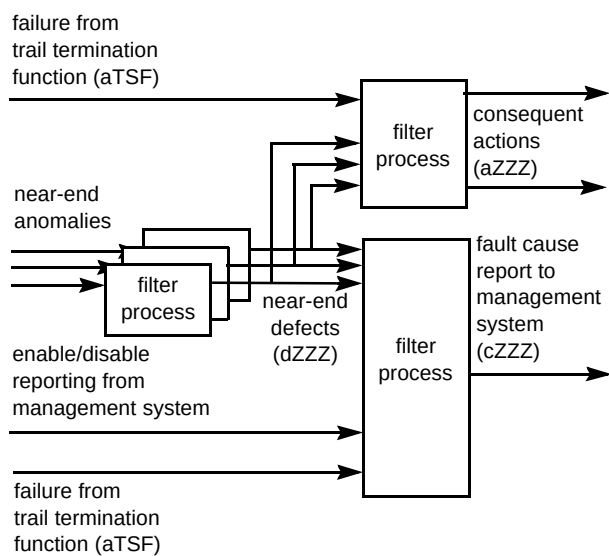


Fig 7 Anomaly processing within adaptation functions.

An anomaly is the smallest deviation from the expected behaviour that can be detected by the equipment. This could, for example, be a single bit error or an error in the received trace identifier string. No action is initiated on the detection of these anomalies but they do initiate further processing to identify defects. A collection of defects may in turn be processed to identify both the cause of the fault and any consequent actions which should be initiated. The processes illustrated in Fig 7 are:

- integration of anomalies into defects,
- consequent action control,
- correlation of defects to identify the fault cause.

The processes are essentially components of the fault management process located within the atomic functions. Other components of the fault management process are located in the equipment management function as illustrated in Fig 5.

7.2 Trail termination functions

According to ITU-T Recommendation G.803, a trail is 'an architectural component in a server layer which transfers client network information transparently between access points at different locations'.

Trail termination functions process the information associated with the creation and monitoring of an SDH trail. The trail termination function performs signal integrity supervision of the layer. In the source direction, it generates and adds the following:

- insertion of any frame alignment word (logically this would be a function of the adaptation function of the layer below, but for historical reasons the process has currently been allocated to the trail termination source function),
- error detection code (e.g. bit interleaved parity, cyclic redundancy check),
- path trace or access point identifier,
- trail signal label (payload composition).

In a bidirectional trail, the source function also adds maintenance signals derived from the associated sink function which reflect the performance of the trail in the opposite direction:

- remote error indicator (REI) containing the number of detected error detection code violations in the received signal,
- remote defect indicator (RDI) representing the defect status of the received signal.

The principle of this far-end monitoring is illustrated in Fig 8.

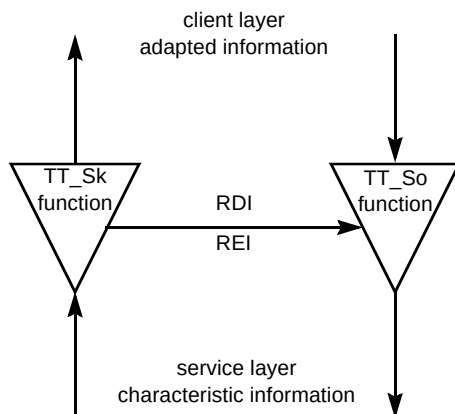


Fig 8 Trail termination function.

In the sink direction, trail termination functions monitor the signal for:

- bit errors,
- (mis)connection,
- (mis)composition of the payload,
- near-end performance,
- far-end performance,
- server signal fail (i.e. AIS instead of data),
- signal loss (disconnection).

Bit errors are detectable via line code violations, parity violations or CRC violations; these are generically referred to as error detection code violations (EDCV).

Access points are identified by means of a fifteen byte string to monitor the provisioning of flexibility within an SDH network. The identifier is inserted in the signal by the trail termination source function. The trail termination sink function checks the received identifier with the expected one (provisioned by the network manager).

Many SDH layers are able to transport a variety of client signals via different adaptation functions. For example, an SDH VC-4 may carry structured or unstructured 140-Mbit/s PDH signals, lower order SDH signals or ATM cells. To monitor the provisioning process, the source trail termination inserts the appropriate code in the trail signal label. The sink trail termination will check the received number with its own provisioned value.

To enable single-ended maintenance of SDH networks, the defect status and number of error detection code violations detected at the sink trail termination are conveyed back to the source trail termination; the defect status via the RDI signal and the number of error detection code violations via the REI signal.

Each termination sink function monitors the following transmission defects:

- alarm indication signal,
- unequipped VC signal,
- trace identifier mismatch,
- signal label mismatch,
- signal degrade,
- loss of signal,
- remote defect indicator.

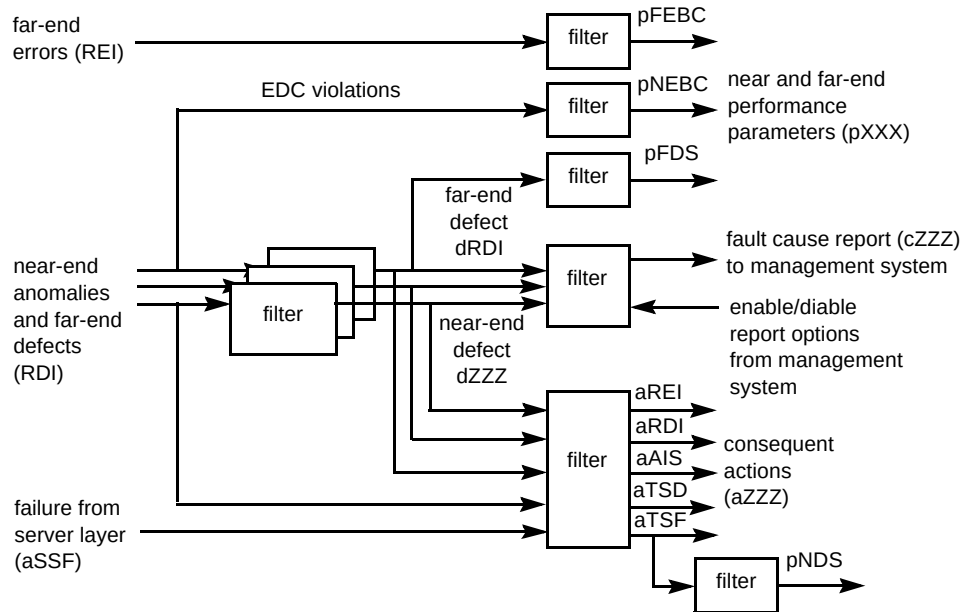


Fig 9 Anomaly processing within trail termination functions.

In the source direction the function monitors for transmitter fail.

As with the adaptation function, anomalies resulting from the detection of these conditions are filtered to identify defects, initiate consequent actions and to identify probable fault causes. However, within trail termination functions, the raw information is also processed to extract trail performance information as illustrated in Fig 9.

Anomalies detected by trail termination functions are processed as described below:

- anomalies are subjected to a check to identify defects (dZZZ),
- certain defects initiate consequent actions (aZZZ),
- defects are correlated to identify the probable fault cause (cZZZ),
- near-end errored blocks are counted (pNEBC),
- near-end defect seconds are counted (pNDS),
- far-end errored blocks are counted (pFEBC),
- far-end defect seconds are counted (pFDS).

Certain processes are components of the fault management process located within the atomic termination functions; others are components of the performance management process located within the atomic termination function. The output signals cZZZ (e.g. cTIM) are the input signals for the equipment management function fault management process, while the output signals pZZZ (e.g. pNEBC) are the input signals for the equipment

management function performance management process, as illustrated in Fig 5.

7.3 Connection functions

The connection function provides flexibility within a network layer. It provides simple connectivity between two or more connection points under the control of management system commands or automatic protection switching protocols.

When the output of a connection function is not connected to a valid input, the connection function automatically inserts the unequipped VC signal.

7.4 Common functions

The equipment management function contains those processes which implement the requirements of the SDH management model. To date, only the fault management and performance management processes have been defined within the ETS.

7.4.1 Equipment management function fault management process

The equipment management function within the network element performs a persistency check on a fault cause before it declares the fault cause to be a failure. The fault causes are generated by the atomic function fault management process.

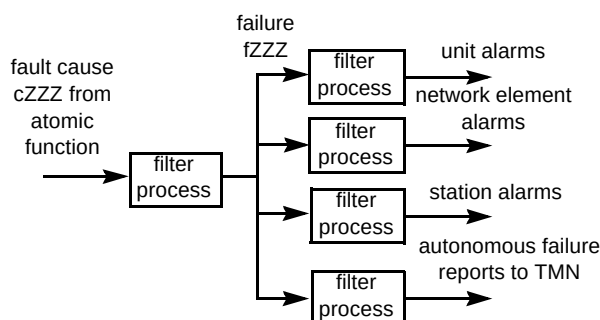


Fig 10 EMF fault management process.

The failure is reported via the agent process and by means of alarms (audible and visible indicators). Typically, alarms can be divided into unit level alarms, network element level alarms and station alarms (see Fig 10).

The fault management processes:

- integrate fault causes to identify failures,
- perform translation of failures into unit alarms,
- perform translation of failures into network element alarms,
- perform translation of failures into station alarms,
- control the selective reporting of faults to the management system.

7.4.2 Equipment management function performance monitoring process

The EMF performance management process (see Fig 11) collects the event counts associated with:

- trail performance parameters,
- link performance parameters,
- protection performance parameters.

It processes the event counts to derive the performance parameters, and stores these parameters in registers. Such information can be used by a network operator to localise faults and to locate sources of intermittent errors, or to determine the quality of the service.

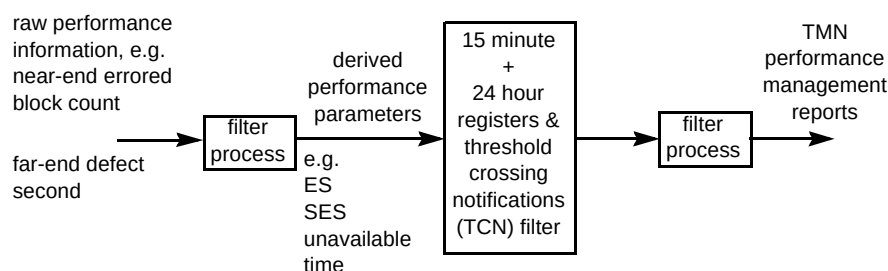


Fig 11 EMF performance management process.

The performance event counts generated by the atomic function performance management process are processed to determine whether a second is an errored second (ES) or a severely errored second (SES), and to determine the number of background block errors (BBE) per second.

The number of these events which occur during a 15 minute and 24 hour period are stored. If the number exceeds a given threshold, an exception report may be generated to the operator.

8. Using the ETS

The intention of the ETS is to describe a number of basic functions which can be connected together to form functional descriptions of SDH equipment. It is intended that in conjunction with a power supply and rack practice statement, such a description would form a full specification for any SDH equipment. In order to fully describe an SDH network element, however, the ETS has provided functions which describe PDH interfaces and path layers. The methodology can therefore be directly applied to describe PDH equipment, and by extension can cover ATM and other technologies.

Functions are combined by binding inputs to outputs at matching reference points to ensure information continuity. A trail termination function that is bound to one of its possible adaptation functions at their common AP, may be extended to form larger equipment descriptions by binding at CPs in a domino-like manner. The process is best illustrated by example. Fig 12 illustrates the functions available in each of the SDH layers.

Figure 12 illustrates the functional components available at each level which are required to describe equipment conforming to the European SDH multiplexing structure. At each level certain functions may be exclusive while others may be added as appropriate. For example, to fully implement the VC4 level conversion, adaptation functions are required to process the multiplex section data communication channel (MS-DCC), the order wire (MS-EOW) and the distribution of synchronisation in addition to any signal conversion. In order to fully describe an add/drop multiplexer, for example, it would be necessary to add higher level SDH and PDH layers and to specify the number of interfaces of each type.

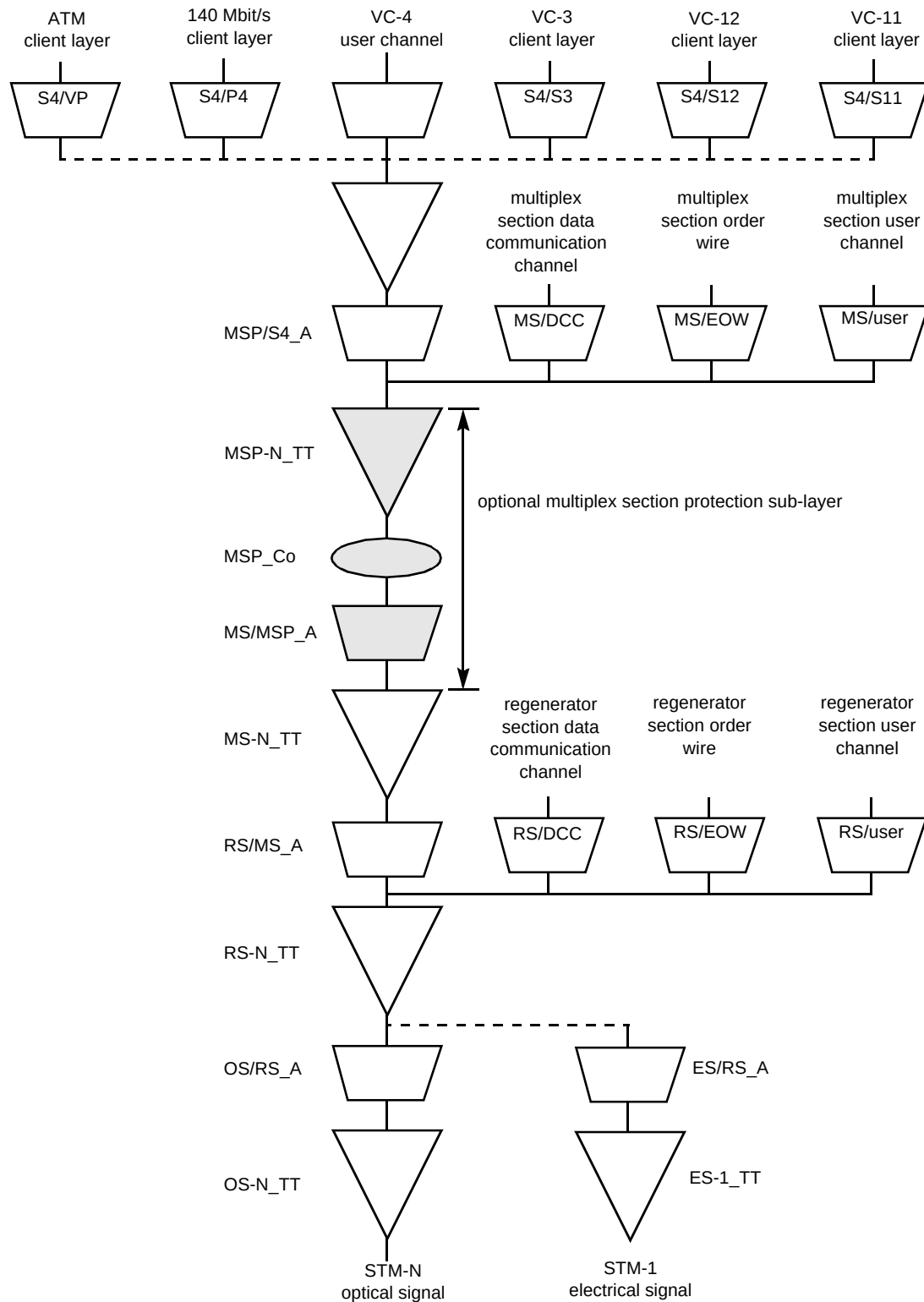


Fig 12 SDH hierarchical level functions.

External interfaces conform to the appropriate international recommendations, but within the functional model only logical reference points are used. The principle is similar, though different, to that employed by the ITU in the production of the SDH equipment recommendation G.783 [3].

8.1 Comparison between ETSI and ITU-T Recommendations

Although ETS 300 417 and G.783 both describe characteristics of SDH equipment functional blocks and both are based upon functional decomposition, comparison

of the two documents is difficult and may be misleading. Consider the functional description of a terminal multiplexer with 2 Mbit/s ports.

Figure 13 illustrates a functional description conforming to ETS 300 417 of an SDH terminal multiplexer excluding

the associated synchronisation. The description includes non-intrusive monitoring of through paths and multiplex section protection.

This can be compared with a description of the same multiplex using the notation described in ITU-T Recommendation G.783 illustrated in Fig 14.

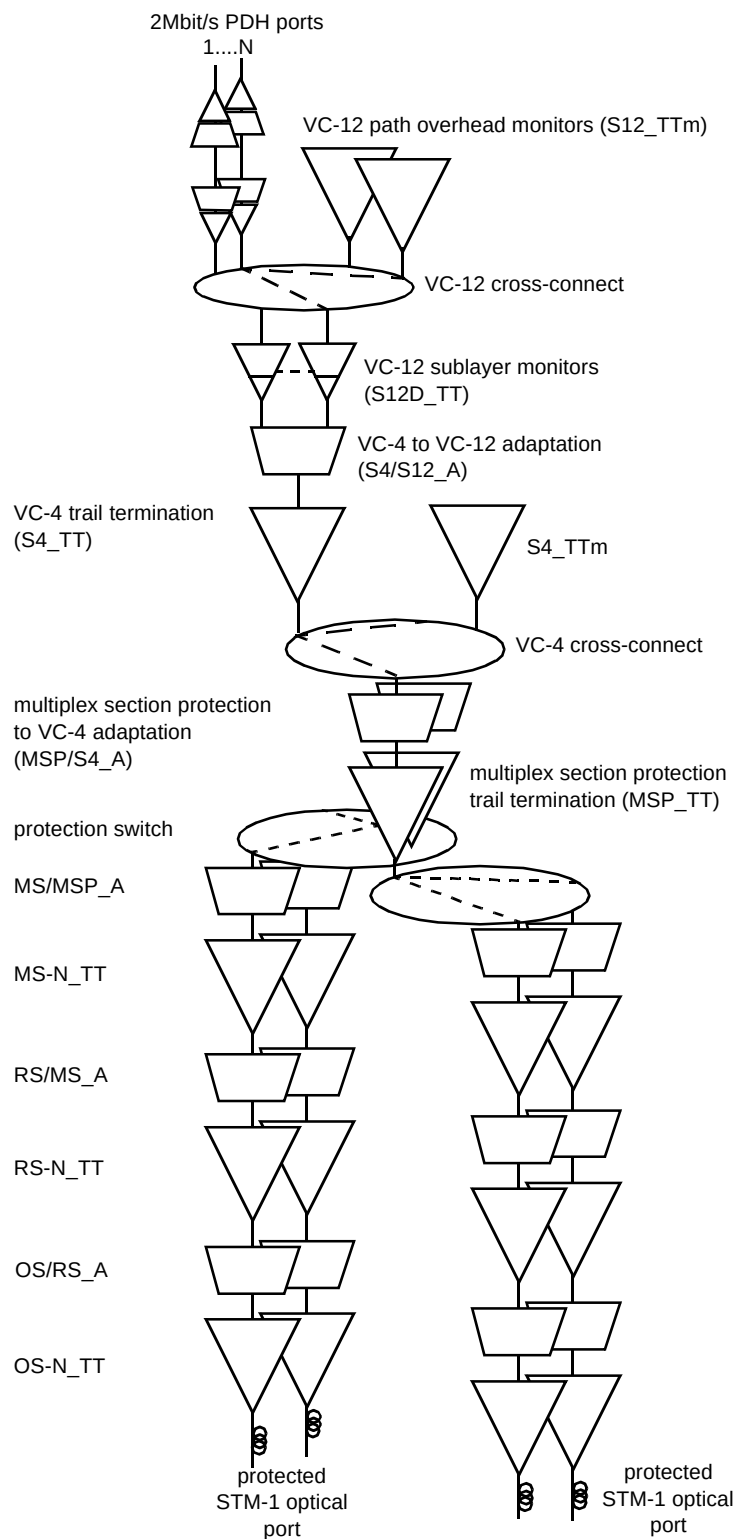


Fig 13 SDH terminal multiplex — ETSI description.

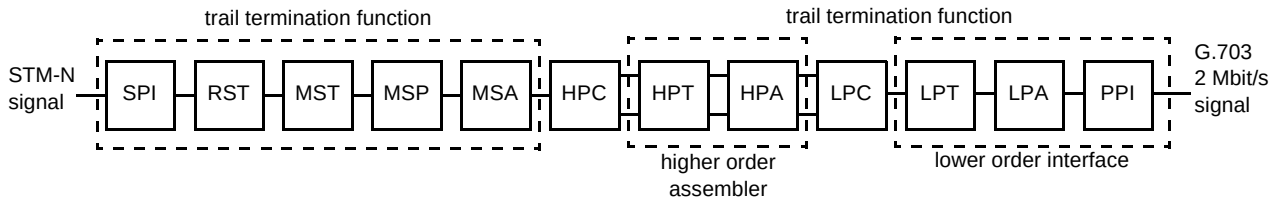


Fig 14 SDH terminal multiplex — ITU description.

The SPI function performs the SDH interface functions of laser control, alarms, optical detection, and timing extraction, and is broadly equivalent to the ETSI OS_TT (which performs signal recovery) and the OS/RS_A (which performs timing recovery) functions. However, the OS/RS_A function also performs frame alignment which is part of the RST function in G.783.

With this exception, the ETSI RS_TT and RS/MS_A functions, and the MS_TT and MS/VC-4_A functions are broadly equivalent to the RST and MST functions in G.783. The MS/VC-4_A function also performs the mapping between VC-4 and AU-4 and the associated byte interleaving and pointer processing actions which are a part of the MSA block in G.783.

The G.783 MSP function is performed by the MSP sub-layer in ETS descriptions, with the actual switch MSP_C being embedded in this layer.

The above examples illustrate that although there is a relationship between the functional block descriptions in ETS 300 417 and G.783, it is neither direct nor obvious. A comparison of the two diagrams also illustrates the major difference between the two methodologies. At first sight, the G.783 technique appears simpler, but, because the technique is less rigorous, equipment from different manufacturers may be functionally identical yet physically incompatible.

9. Specific application examples

The functional modelling technique may be used to describe transmission equipment or to unambiguously describe specific functions. Examples of these are given in the remainder of this section; all the functionality could be present in a specific equipment.

9.1 Path performance monitoring

SDH equipment provides a variety of functions which may be used to monitor the in-service performance of paths. These are illustrated below by an example of a typical path which originates and terminates on SDH VC-4/1 cross-connects and transits a VC-4/4 crossconnect. The example illustrates a number of monitoring techniques, not all of which would normally be applied to a single path.

The regenerated line termination (RLT) is a compound function combining the optical section, regenerator section and multiplex section layers.

The performance of the client service, in this case a 2-Mbit/s circuit, may be monitored at the output of the equipment through any appropriate technique (e.g. CRC-4) and the performance passed to the management system.

The performance of the VC-12 path across the network may be monitored at the output of the equipment through the VC-12 path overhead and the performance information passed to the management system. The same parameters may be monitored at any intermediate nodes having visibility of the path through the application of non-intrusive monitors (VC-12_TTm in Fig 15).

Parameters extracted from each node may be correlated at the management system level to isolate link performance. In cases where the management system is incomplete, the link information may be correlated through sub-layer tandem connection monitoring techniques which measure far-end error performance and return the information via an embedded data link (VC-12 TCM in Fig 15).

The performance of a client layer can be estimated from that of its server layer if certain assumptions concerning error distribution are made.

9.2 Protection

All types of protection can be classified as either trail protection or sub-network connection protection. These are described below.

9.2.1 Trail protection

In this scheme, a working trail is replaced by a protection trail if the working trail fails or if the performance falls below the required level. The process is implemented by including a protection sub-layer between the trail termination and the adaptation to the next layer, i.e. at the access point within the protected layer. The most obvious application of trail protection is multiplex section protection (MSP). This is illustrated in Fig 16 for the specific case of one-plus-one protection.

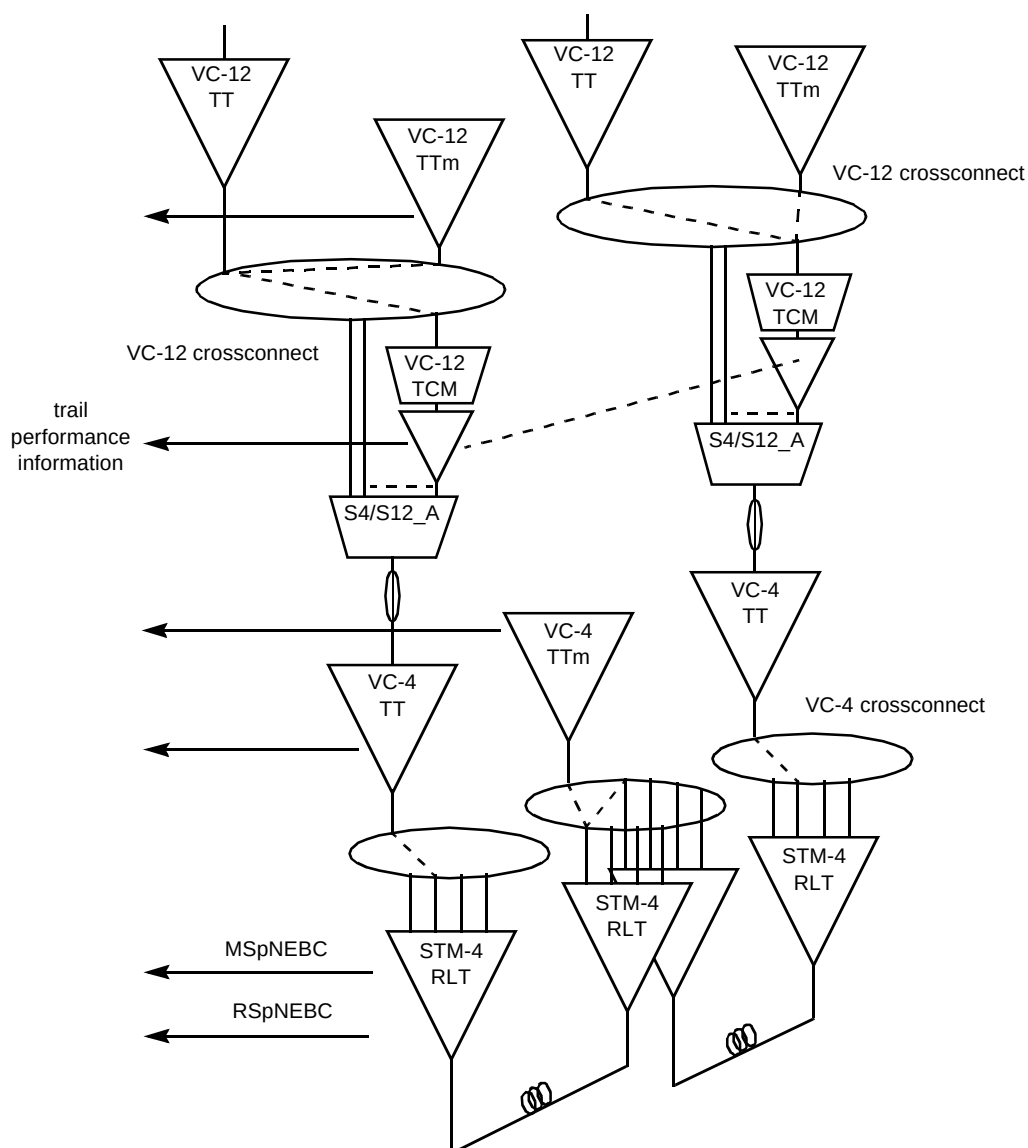


Fig 15 Path performance monitoring model.

The status of the two trails is made available to the switching function through the trail signal fail (TSF) and the protection adaptation function provides access to an automatic protection switching (APS) protocol. MSP provides a fast (50 ms) means of restoring capacity in the event of a failure.

9.2.2 Sub-network connection protection

In this scheme, a working sub-network connection is replaced by a protection sub-network connection if the working sub-network connection fails or if the performance falls below the required level. The protection may be applied at any level and is implemented by including a sub-layer between layers in order to monitor the sub-network connections, i.e. at the connection point. The most obvious application of sub-network connection protection is ring protection. This is illustrated in Fig 17.

The status of the sub-network connections is monitored by the new (sub-) layer which passes information on server signal fail (SSF) to the protection switch. Monitoring may be implemented using any of the schemes described.

The activation time of sub-network connection protection has not yet been agreed, but it will generally be slower than that provided by trail protection in order to avoid contention effects.

10. Benefits

The functional decomposition technique offers benefits to both network operators and equipment manufacturers.

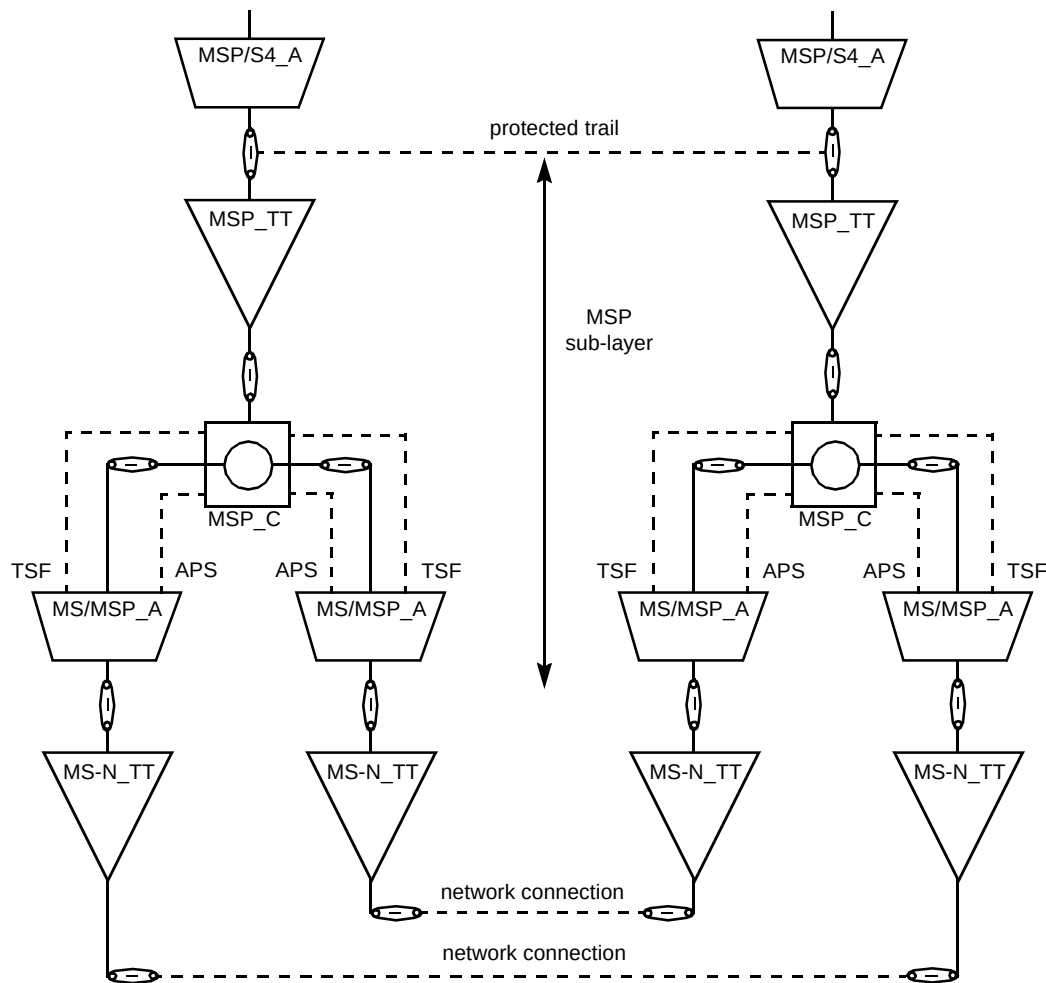


Fig 16 Trail protection functions.

- Manufacturers

The main benefit for suppliers is that all equipment purchasers who adopt the methodology of the ETS will use a common description. Hence manufacturers will no longer be forced to invest resources interpreting the requirements of individual customers.

- Operators

The main benefit to operators of the technique employed in ETS 300 417 is to remove the need for international standards to be agreed for each type of equipment in the network. Once agreed, the functions will allow an operator to explicitly specify an SDH line system, add/drop multiplex or crossconnect. As many of the functions employed in SDH equipment are common to other equipments, they may be used to specify components of ATM, PDH or FMUX equipment without specific standardisation of these equipments.

Another benefit is to remove ambiguity and hence to ensure interoperability between equipment. The functional description of equipment contained in ITU-

T Recommendation G.783 allows various options in each block which prevent interoperability being guaranteed. ETS 300 417 removes this ambiguity — where options exist they are specified as separate atomic functions.

11. Further development

ETS 300 417 provides a means to unambiguously describe the functionality of telecommunications equipment by the use of a modelling technique through which the external behaviour of an equipment is attributed to a connected set of 'atomic' functions.

The complete set of atomic functions may be interconnected to model the behaviour of any transmission equipment. To date, the parts of the ETS which describe general behaviour and the principles of the technique, and SDH path and section layer functions, have been approved within ETSI. Further parts to describe the atomic functions in the PDH path layers and synchronisation layer are now being developed.

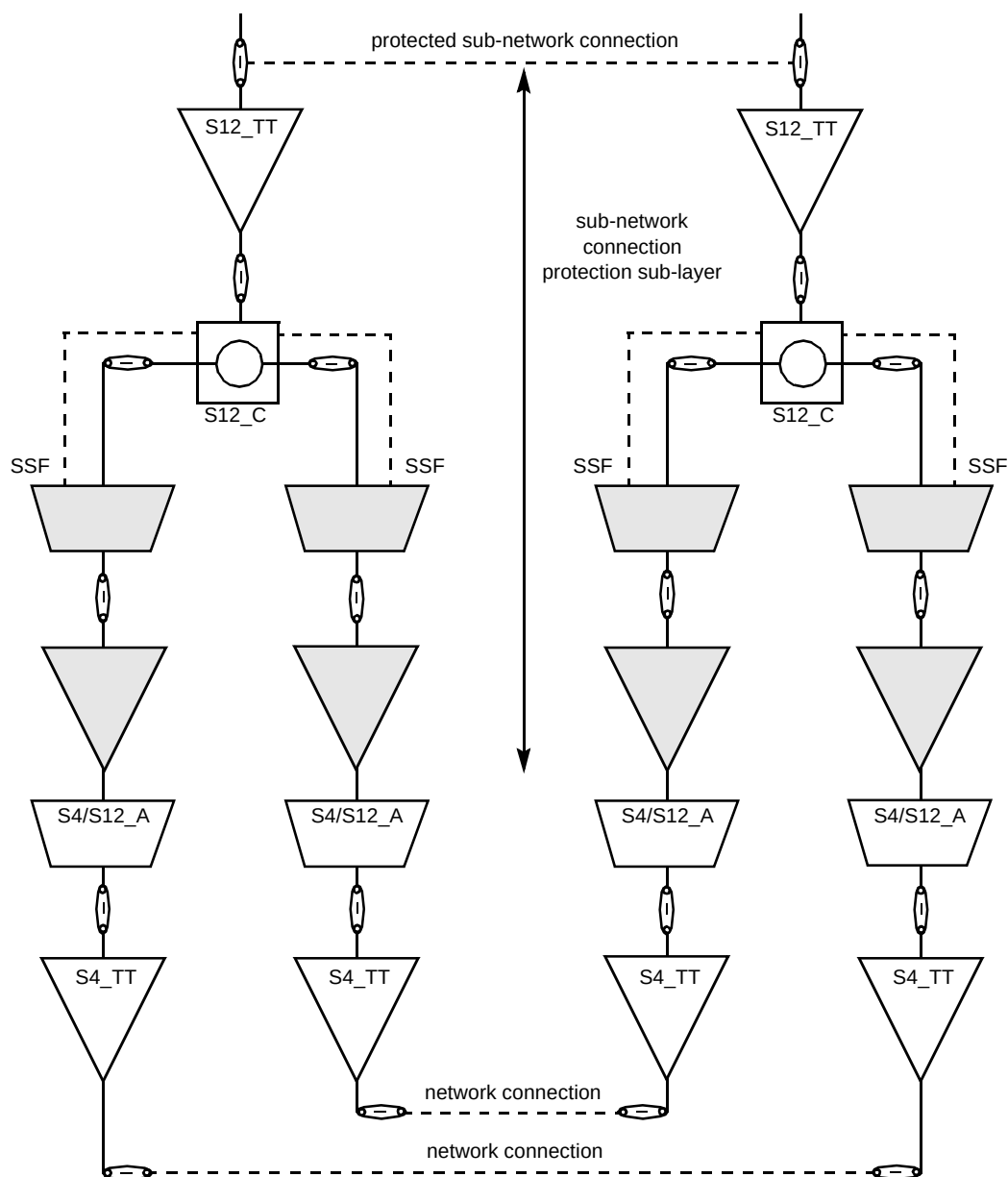


Fig 17 SNC protection functions.

The technique is not only applicable to SDH equipment. Groups within ETSI TM1 are currently applying the principles to describe flexible multiplexer and ATM equipments.

12. Conclusions

ETS 300 417 is designed to provide a library of functions which may be combined to provide functional descriptions of SDH equipment. Such a description will fully describe the performance, defect handling and functionality of any such equipment.

The ETS which describes generic processes and performance, and which provides an introduction to the technique has been approved by ETSI and published. Other

parts which describe individual atomic functions are currently being produced.

Acknowledgements

The material contained in this paper is based on current versions of the ETS and contributions by the author to the development process. The author fully acknowledges the contribution of many other individuals throughout Europe to the development of the standard and intellectual ownership of specific material referred to herein. The author offers his grateful thanks to his colleagues within the standards community for a number of useful discussions of the material contained in this paper.

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